MLOps Exposé — Docker

1. Presentation

Docker is an open-source platform that allows developers to build, package, ship, and run applications inside lightweight, portable containers. Containers include everything the application needs, ensuring portability, consistency, speed, and isolation.

2. Main Features

- Containerization Engine: Runs apps in isolated environments.

- Docker Images: Reusable templates containing code and dependencies.

- Docker Containers: Fast, isolated runtime instances.

- Docker Hub: Repository for downloading and sharing images.

- Docker Compose: Manages multi-container applications.

- Portability Across Clouds.

3. Type / Cost

Type: Containerization and DevOps tool.

Cost: Free for individuals, students, and basic use. Paid plans available but not necessary for student work.

4. Stage in DevOps Process

- Plan: Define architecture using container services.

- Build: Create Docker images.

- Test: Run automated tests inside containers.

- Release: Push/pull images using CI/CD.

- Deploy: Deploy containers to cloud/Kubernetes.

- Operate: Run applications in production.

- Monitor: Use tools like Prometheus/Grafana.

5. Advantages and Limits

Advantages:

- Highly portable
- Lightweight
- Fast deployment
- Reproducible environments
- Huge ecosystem

Limits:

- Learning curve
- Not a full OS virtualization
- Docker Desktop can be heavy
- Needs Kubernetes for large-scale systems

6. Field Applications

- Machine Learning and MLOps
- Web development
- Microservices architectures
- Cloud computing (AWS, GCP, Azure)
- Education and research

7. Conclusion

Docker revolutionized deployment by enabling fast, isolated, and portable software environments. It is a core DevOps and MLOps tool widely used in industry and academia.